



IILM University

GREATER NOIDA · U.P.

School of Computer Science
& Engineering

INTERNSHIP PRESENTATION · 2026

DATA SCIENCE & MACHINE LEARNING INTERNSHIP

Applying Community Detection to Find Habitat Clusters

A graph-based ecological analysis project that represents habitat locations as a network and uses community detection to identify naturally connected habitat clusters.

STUDENT	PROGRAM	ORGANIZATION
Sachin Kumar 25SCS1003004017	B.Tech CSE 3rd Semester · Batch 2025–2029	upSkill Data Science Intern · 25 Jul – 25 Aug 2026

Where I applied data science to an ecological problem.

A one-month online Data Science & Machine Learning internship focused on practical data preparation, analysis, modeling concepts, evaluation, and project-based implementation.

01 · ORGANIZATION



For this presentation, **upSkill** is used as the internship organization label, while the project is presented within the Data Science & Machine Learning domain. The work emphasizes a practical workflow from data preparation to analysis, interpretation, and project documentation.

Data Science · Graph Analytics · Machine Learning

02 · THE PROGRAM

Data Science Internship

ROLE

Data Science Intern

DURATION

1 Month · 25 Jul – 25 Aug 2026

FOCUS

Data preparation, graph construction, community detection, visualization and interpretation.

What I built — and why.

Habitats are not isolated units: they can be connected by geographic proximity, ecological similarity, migration pathways, or shared environmental conditions. A graph model can expose these relationships and reveal communities that may be missed by simple grouping.

PROBLEM STATEMENT

01 · PROJECT OBJECTIVES

- 01 Represent habitat locations as nodes in a graph.
- 02 Create weighted edges from spatial or ecological similarity.
- 03 Apply community detection to identify dense habitat communities.
- 04 Visualize the detected clusters for interpretation.
- 05 Evaluate community structure using modularity and graph measures.
- 06 Document the workflow for reproducible ecological analysis.

02 · SCOPE

Graph-based habitat analysis

Scope covers data cleaning, feature preparation, habitat-network construction, community detection, cluster visualization, and interpretation. It does not attempt to perform species classification, satellite-image segmentation, live tracking, or a production conservation-management system.

Stack & system architecture.

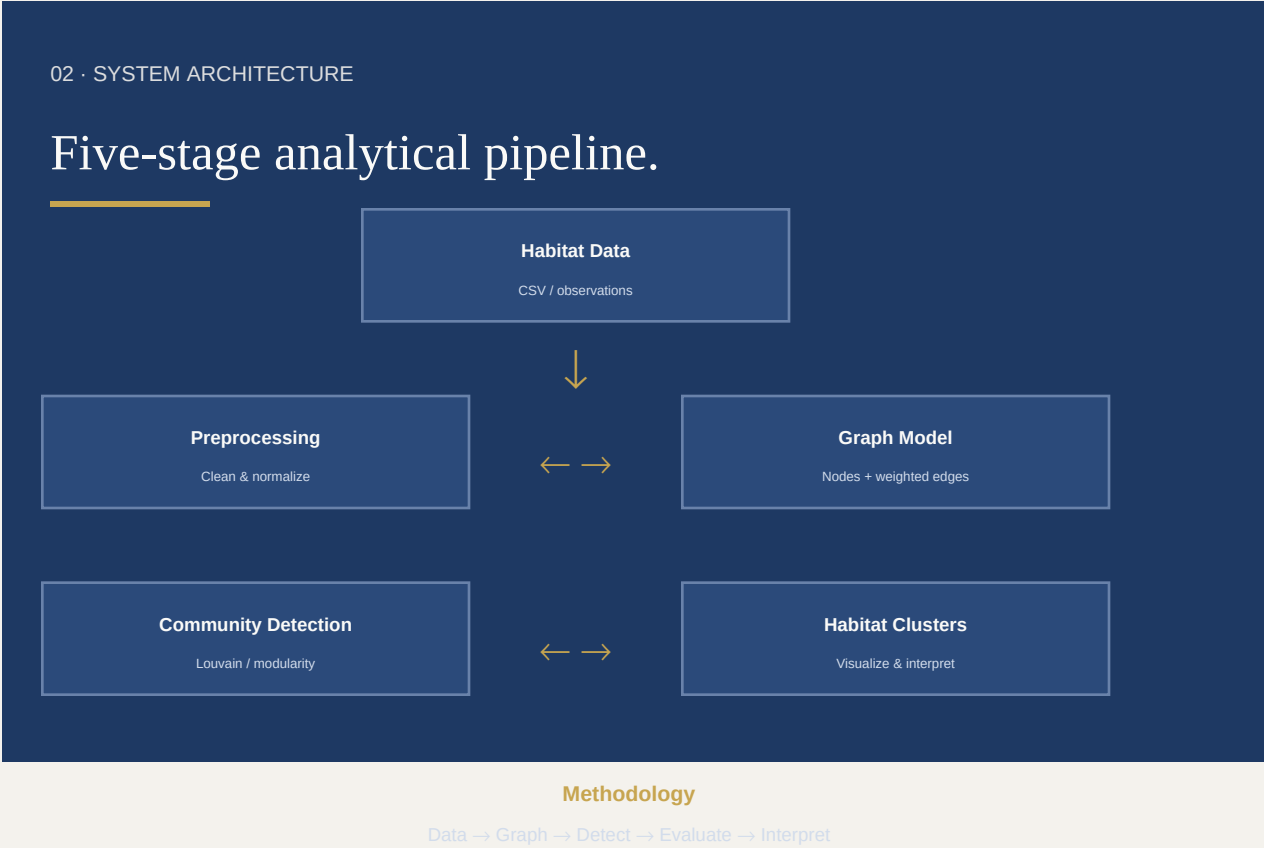
A compact analytical pipeline turns habitat observations into a graph, detects communities, evaluates the structure, and produces an interpretable cluster map.

01 · TECH STACK

Major technologies & tools.

Programming	Python	Data processing & analysis
Libraries	Pandas, NumPy	Cleaning & transformation
Graph Analysis	NetworkX	Graph & communities
Visualization	Matplotlib	Cluster visualization
Environment	Jupyter Notebook	EDA & experiments
Version Control	Git & GitHub	Project management

Table 4.1 – Technologies Used



What I take away.

The project demonstrates how graph-based community detection can turn habitat relationships into meaningful clusters for analysis and visualization.

01

Habitat network constructed from ecological relationships.

02

Weighted graph used to represent habitat connectivity.

03

Community detection applied to identify habitat clusters.

04

Cluster structure visualized for easier interpretation.

05

Hands-on experience with Python, NetworkX, Pandas, visualization and graph analytics.

